



# AI Evaluation Report

Data Science Expert — Self-Assessment

CANDIDATE

Ritam

FINAL SCORE

24 / 70

weighted points

CORRECT ANSWERS

10 / 35

questions

NEEDS IMPROVEMENT

Assessment Date: July 25, 2026

Report Generated: July 25, 2026 at 10:39 UTC

Role: Data Science Expert

OVERALL RATING

★★★★★

SUCCESS RATE

34.3%

WEIGHTED SCORE

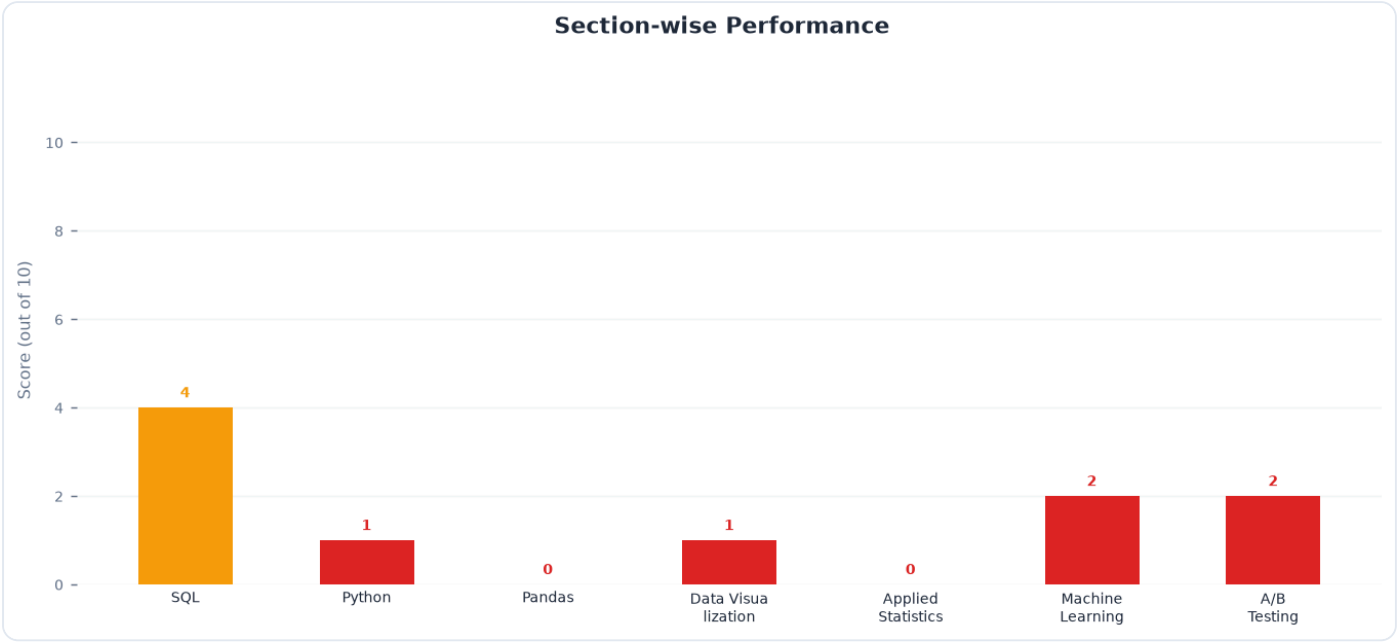
24 / 70

TOP SKILL

SQL

NEEDS IMMEDIATE ATTENTION

Pandas



## Skill Rating & Benchmark

| Section            | Your Score | Reference Benchmark (Demo) | Delta | Status       |
|--------------------|------------|----------------------------|-------|--------------|
| SQL                | 4 / 10     | 6.8                        | -2.8  | Needs Work   |
| Python             | 1 / 10     | 6.2                        | -5.2  | Critical Gap |
| Pandas             | 0 / 10     | 5.8                        | -5.8  | Critical Gap |
| Data Visualization | 1 / 10     | 6.0                        | -5.0  | Critical Gap |
| Applied Statistics | 0 / 10     | 5.5                        | -5.5  | Critical Gap |
| Machine Learning   | 2 / 10     | 5.2                        | -3.2  | Needs Work   |
| A/B Testing        | 2 / 10     | 6.4                        | -4.4  | Critical Gap |

Attempt Statistics Breakdown

| Section            | Total Q | Attempted | Correct | Incorrect | Skipped | Accuracy |
|--------------------|---------|-----------|---------|-----------|---------|----------|
| SQL                | 5       | 5         | 4       | 1         | 0       | 80%      |
| Python             | 5       | 5         | 1       | 4         | 0       | 20%      |
| Pandas             | 5       | 5         | 0       | 5         | 0       | 0%       |
| Data Visualization | 5       | 5         | 1       | 4         | 0       | 20%      |
| Applied Statistics | 5       | 5         | 0       | 5         | 0       | 0%       |
| Machine Learning   | 5       | 5         | 2       | 3         | 0       | 40%      |
| A/B Testing        | 5       | 5         | 2       | 3         | 0       | 40%      |

Executive Summary

The candidate achieved a score of 10 out of 35 correct answers, resulting in a weighted score of 24 out of 70. This performance indicates significant gaps in core data science competencies, particularly in Python/Pandas execution, statistical theory, and advanced experimentation. Systematic upskilling is required to transition from basic conceptual familiarity to production-level execution.

Key Strengths

- Basic SQL & Python Familiarity:** Demonstrates foundational exposure to data manipulation syntax and basic programming structures.
- Broad Domain Exposure:** Shows awareness of advanced topics like CUPED, FDR control, and learning curves, even though technical execution was incorrect.

Top Blind Spots

- **Python Closures and Pandas Mechanics:** The candidate struggles with core programming concepts like closures in decorators and essential Pandas operations, such as index monotonicity in rolling windows and proper `.rename()` syntax.
- **Statistical Inference and Hypothesis Testing:** There is a fundamental misunderstanding of the Central Limit Theorem, Q-Q plot interpretations, and the mechanics of False Discovery Rate control (Benjamini-Hochberg).
- **Experimentation and Advanced A/B Testing:** The candidate lacks technical clarity on variance reduction techniques like CUPED and sequential testing boundaries (e.g., O'Brien-Fleming).

Actionable Learning Resources

**Pandas & Python: Complete the "Pandas Foundations" course on DataCamp, focusing on grouping, multi-indexing, and window operations.**

✓ Action Item

**Applied Statistics: Read Chapter 2 (Probability) and Chapter 3 (Inference) of \*OpenIntro Statistics\*, and practice Q-Q plot analysis.**

✓ Action Item

**A/B Testing: Read "Trustworthy Online Controlled Experiments" (Kohavi et al.), specifically Chapter 16 on variance reduction (CUPED) and Chapter 18 on sequential testing.**

✓ Action Item

**Machine Learning: Study Chapter 2 of \*Introduction to Statistical Learning (ISLR)\* to master bias-variance trade-offs and learning curve diagnostics.**

✓ Action Item